MARÍA JOSÉ BUSTAMANTE ROSELL

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POSITIONS HELD

Scientist *March 2026-Present*
Universities Space Research Association (USRA)

EMIT Postdoctoral Fellow *2024-2026*
Fisk and Vanderbilt Universities, Nashville

Postdoctoral Scholar *2021-2024*
University of California, Santa Cruz

EDUCATION

The University of Texas at Austin *2014-2021*
PhD in Physics, Advisor: Aaron Zimmerman

Pontificia Universidad Católica del Perú (PUCP) *2008-2013*
BSc in Physics (first in class), 2012

RESEARCH FUNDING SUMMARY

Total competitive funding secured as PI: $588,370

SPACE & OBSERVATORY PROGRAMS

PI — JWST Cycle 3 Archival Research Program ID 5355 *2025–2028*
Space Telescope Science Institute / JWST (NASA/ESA). Award: $220,982. Program page.
Title: Adapting the Constrained Matrix Factorization Deblender `Scarlet` to Separate Overlapping Sources in JWST IFU Data.

Co-I — JWST General Observer Program ID 3805 (Cycle 2) *2023–2025*
Space Telescope Science Institute / JWST (NASA/ESA). Program page.
Contribution: Proposal design, observing plan, analysis strategy.

NASA LISA PREPARATORY SCIENCE (LPS)

PI — Electromagnetically-Driven GW Searches for EMRIs *2025–2028*
Proposal 24-LPS24-0023. Award: $367,388. Multi-messenger Bayesian framework linking EM timing (QPE/QPO) with GW morphology for "wet" EMRIs.

Co-I — A Robust Search for Faint and Weakly Modeled Sources in LISA *2024*
Semi-coherent, weakly modeled pipeline; chirplet banks + HMM/PCA; *includes GPU-accelerated template evaluation.* NASA panel rating: **Very Good**.

Science PI — LISA Sources in the Galactic Center (Proposal 22-LPS22-0035) *2022*
Tidally-aware X-MRI waveform development; GLASS integration; NASA panel rating: **Excellent / Very Good**.

COLLABORATIONS

LISA Consortium *2021-Present*
Young Supernova Experiment (YSE) *2021-2023*
LIGO-HET Response (LIGHETR)
Electromagnetic Counterpart Search **Team Leader** *2019-2021*
SXS Collaboration *2018-2021*
LIGO/Virgo Scientific Collaboration *2018-2021*
MEDIDO PROJECT *2016-2018*
MINERvA *2013-2014*

FELLOWSHIPS AND AWARDS

Establishing Multimessenger Astronomy Interdisciplinary Training (EMIT)
Postdoctoral Fellowship *Full financial support* *2024-2027*

Good Neighbour Scholarship *Full year tuition* *2017-2018*

John A. Wheeler Fellowship in Physics *Summer tuition and stipend* *Summer 2016*

Full Scholarship for pursuing the MSc in Physics at PUCP. *2012*

MENTORSHIP AND LEADERSHIP

Founder & Director — Fisk–Vanderbilt IDEA Summer Scholars Program *2023–Present*
Created a fully funded summer program pairing three Fisk data-science undergraduates with Vanderbilt postdocs on astrophysics/data-science projects (selection, matching, funding).

TEACHING AND MENTORING

Co-Director, IDEA Summer Scholars Program (Fisk–Vanderbilt) *2024–Present*
Designed and co-led a bridge research and training program for undergraduates from HBCUs entering astrophysics. Supervised 6 mentees; coordinated joint mentoring with Vanderbilt and UT Austin faculty.

Instructor / Teaching Assistant, University of Texas at Austin *2014–2019*
Led discussion and lab sections for Classical Mechanics, Introductory Astronomy, and Computational Physics. Developed assessment rubrics and mentoring strategies for diverse student backgrounds.

Graduate Mentor, Vanderbilt *2025–2024*
Mentored one graduate student in gravitational-wave data analysis and high-performance computing.
Undergraduate Mentor,

TEACHING EXPERIENCE

UT-Austin *2014-2021*

- TA for General Relativity
- AI for Physical Science 303 (Instructor of record for the class)
- TA for Popular Astronomy
- TA for General Physics I

PUCP *2011-2012*

- TA for Cosmology
- TA for Introduction to Research

SELECTED TALKS AND COLLOQUIA

Texas Tech University Astronomy Colloquium

Black Holes and How to Find Them: Correlation-Driven Multimessenger Inference in Messy Astrophysics *Lubbock* *February 19 2026*

Duke University Cosmology Seminar

Science with the millihertz Gravitational Wave Sky: Gravitational Wave Timing Arrays and Electromagnetically Driven EMRI Searches *Durham* *April 4 2025*

University of Washington Astronomy Colloquium *Seattle* *February 15 2024*

Black Holes and How to Find Them

Space Science Seminar *NASA Marshall* *March 7 2023*

Kinematic Constraints on Supermassive Black Holes in the Local Group and Beyond

Rochester Institute of Technology Astronomy Colloquium *Rochester* *December 5 2022*

The supermassive black hole at the center of Leo I

Lisa Symposium XIV *Virtual* *July 25 2022*

Gravitational Wave Timing Array

Intermediate-Mass Black Holes Conference *Puerto Rico* *March 1 2022*

Invited Talk - Dynamical analysis of the matter content in the Dwarf Spheroidal Galaxy Leo I

APS April Meeting 2020 *Virtual*

The LIGO HET Response (LIGHETR) Project to Discover and Spectroscopically Follow Optical Transients Associated with Neutron Star Mergers

CENTRA seminar series *Lisbon* *Dec 10 2020*

Self-torque and frame nutation in binary black hole simulations

DESY Theoretical BS Astroparticle Seminar, DESY Zeuthen, Germany *Mar 17 2020*

Brown Bag Seminar, The University of Texas at Austin, *Austin*

Black Hole Spin Precession in the Self-Force Approximation

APS April Meeting 2019 *Colorado*

Dynamical analysis of the dark matter and black hole mass in the dwarf spheroidal LEO I

EXGal Seminar, The University of Texas at Austin, *Austin*

Dark matter density profiles in local dSphs: *November 30, 2017*

The Leo I case

OPINAS Seminar, Max-Planck-Institut für extraterrestrische Physik, *Garching*

LEO I'S CENTRAL POTENTIAL: *July 31, 2017*

Measurement difficulties and estimations from dynamical models

OPINAS Ringberg Meeting, Max-Planck-Institut für extraterrestrische Physik, *Schloss Ringberg*

Central Dark Matter Density Profiles *April 6, 2017*

LEO I

PUBLICATIONS

[1] D. A. Coulter et al. The Gravity Collective: A Comprehensive Analysis of the Electromagnetic Search for the Binary Neutron Star Merger GW190425. *Astrophys. J.*, 988(2):169, 2025.

[2] D. Farias, C. Gall, V. A. Villar, K. Auchettl, K. M. de Soto, A. Gagliano, W. B. Hoogendam, G. Narayan, A. Sedgewick, S. K. Yadavalli, Y. Zenati, C. R. Angus, K. W. Davis, J. Hjorth, W. V. Jacobson-Galán, D. O. Jones, C. D. Kilpatrick, M. J. Bustamante Rosell, D. A. Coulter, G. Dimitriadis, R. J. Foley, et al. Characterization of type Ibn SNe. 11 2025.

[3] W. V. Jacobson-Galán et al. A Panchromatic View of Late-time Shock Power in the Type II Supernova 2023ixf. *Astrophys. J. Lett.*, 2025.

[4] R. Abbott et al. GWTC-2.1: Deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run. *Phys. Rev. D*, 109(2):022001, 2024.

[5] E. Padilla Gonzalez et al. SN 2022joj: A Potential Double Detonation with a Thin Helium Shell. *Astrophys. J.*, 964(2):196, 2024.

[6] Lindsey A. Kwok et al. Ground-based and JWST Observations of SN 2022pul. II. Evidence from Nebular Spectroscopy for a Violent Merger in a Peculiar Type Ia Supernova. *Astrophys. J.*, 966(1):135, 2024.

[7] Matthew R. Siebert et al. Ground-based and JWST Observations of SN 2022pul. I. Unusual Signatures of Carbon, Oxygen, and Circumstellar Interaction in a Peculiar Type Ia Supernova. *Astrophys. J.*, 960(1):88, 2024.

[8] S. Tinyanont et al. Keck Infrared Transient Survey. I. Survey Description and Data Release 1. *Publ. Astron. Soc. Pac.*, 136(1):014201, 2024.

[9] K. W. Davis et al. SN 2022ann: a Type Icn supernova from a dwarf galaxy that reveals helium in its circumstellar environment. *MNRAS*, 523(2):2530–2550, August 2023.

[10] P. D. Aleo et al. The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae. *Astrophys. J. Suppl.*, 266(1):9, 2023.

[11] M. J. Bustamante-Rosell et al. The LIGO HET Response (LIGHETR) Project to Discover and Spectroscopically Follow Optical Transients Associated with Neutron Star Mergers*. *Astrophys. J.*, 954(1):102, 2023.

[12] D. A. Coulter et al. YSE-PZ: A Transient Survey Management Platform that Empowers the Human-in-the-loop. *Publ. Astron. Soc. Pac.*, 135(1048):064501, 2023.

[13] K. W. Davis et al. SN 2022ann: a Type Icn supernova from a dwarf galaxy that reveals helium in its circumstellar environment. *Mon. Not. Roy. Astron. Soc.*, 523(2):2530–2550, 2023.

[14] Jean-Baptiste Fouvry, María José Bustamante-Rosell, and Aaron Zimmerman. Constraining intermediate-mass black holes from the stellar disc of SgrA*. *Mon. Not. Roy. Astron. Soc.*, 526(1):1471–1481, 2023.

[15] Peyton T. Johnson, Michael W. Coughlin, Ashlie Hamilton, María José Bustamante-Rosell, Gregory Ashton, Samuel Corey, Thomas Kupfer, Tyson B. Littenberg, Draco Reed, and Aaron Zimmerman. Multimessenger parameter inference of gravitational-wave and electromagnetic observations of white dwarf binaries. *Mon. Not. Roy. Astron. Soc.*, 525(3):4121–4128, 2023.

[16] María José Bustamante-Rosell, Joel Meyers, Noah Pearson, Cynthia Trendafilova, and Aaron Zimmerman. Gravitational wave timing array. *Phys. Rev. D*, 105(4):044005, 2022.

[17] John A. Regan, Fabio Pacucci, and M. J. Bustamante-Rosell. Observational signatures of massive black hole progenitor pathways: could Leo I be a smoking gun? *Mon. Not. Roy. Astron. Soc.*, 518(4):5997–6003, 2022.

[18] R. Abbott et al. GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run. *Phys. Rev. X*, 11:021053, 2021.

[19] R. Abbott et al. Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog. *Astrophys. J. Lett.*, 913(1):L7, 2021.

[20] Maria Jose Bustamante-Rosell, Eva Noyola, Karl Gebhardt, Maximilian H. Fabricius, Ximena Mazzalay, Jens Thomas, and Greg Zeimann. Dynamical Analysis of the Dark Matter and Central Black Hole Mass in the Dwarf Spheroidal Leo I. *Astrophys. J.*, 921(2):107, 2021.

[21] Alejandro Torres-Orjuela, Pau Amaro Seoane, Zeyuan Xuan, Alvin J. K. Chua, María J. B. Rosell, and Xian Chen. Exciting Modes due to the Aberration of Gravitational Waves: Measurability for Extreme-Mass-Ratio Inspirals. *Phys. Rev. Lett.*, 127(4):041102, 2021.

[22] B. P. Abbott et al. GW190425: Observation of a Compact Binary Coalescence with Total Mass $\sim 3{,}4 M_{\odot}$. *Astrophys. J. Lett.*, 892(1):L3, 2020.

[23] B. P. Abbott et al. GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs. *Phys. Rev. X*, 9(3):031040, 2019.

[24] B. P. Abbott et al. Tests of General Relativity with the Binary Black Hole Signals from the LIGO-Virgo Catalog GWTC-1. *Phys. Rev. D*, 100(10):104036, 2019.

[25] B. Eberly+ et al. Charged pion production in ν_μ interactions on hydrocarbon at $\langle E_\nu \rangle = 4{,}0\,\mathrm{GeV}$. *Phys. Rev. D*, 92:092008, Nov 2015.

[26] T. Walton et al. Measurement of muon plus proton final states in ν_μ interactions on hydrocarbon at $< E_\nu >= 4.2$ GeV. *Phys. Rev.*, D91(7):071301, 2015.

[27] B. G. Tice et al. Measurement of Ratios of ν_μ Charged-Current Cross Sections on C, Fe, and Pb to CH at Neutrino Energies 2-20 GeV. *Phys. Rev. Lett.*, 112(23):231801, 2014.

INVITATIONS AND VISITS

Visiting Scientist at Max-Planck-Institut für extraterrestrische Physik (Medido Project) *2017*

- Collaborated on the kinematic measurements and dynamical models for

Dwarf Spheroidals as part of the Medido Project.

- Planning of observations of ultra faint Dwarf Spheroidals with VIRUS-W.

Visiting Scientist at Fermilab (MINERvA) *2014-2015*

- Leaded the implemention of the fourth Muon Monitor in the NuMI beamline.

- Studied and calibrated the flux pipeline in the NuMI beamline.

- Work on the study and design of new Cherenkov detectors for DUNE.

PROFESSIONAL SERVICE

Scientific Organizing Committee: LISA Symposium *2026*

Proposal Reviewer: NASA Euclid General Investigator Program (EGIP) *2025*

Local Organizer: 3rd LISA Sprint, Huntsville, AL *2025*

Peer Reviewer: Nature Astronomy *2023*

ACTIVITIES

Research Assistant for LIGO and SXS collaborations	*2018-2021*
Research Assistant for HETDEX collaboration	*2015-2021*
Research Assistant in PUCP's High Energy Physics Group	*2013-2014*
Research Assistant in PUCP's Quantum Optics Group	*2010-2012*
Volunteer Teacher for NGO "Bruce Peru"	*2010-2012*
Research Assistant for NGO Otra Mirada	*2009*
Development and Implementation of Online Queries – Overseas Development Institute (ODI)	*2008*
Member of PUCP's Complex Systems Investigation Group	*2007-2012*
Foundational Member of Astronomy Group in PUCP	*2006*
Volunteer Firefighter	*2004-2006*

OUTREACH

Fisk-Vanderbilt Interdisciplinary Data-science Experience in Astrophysics (IDEA) Summer Scholarship – *Nashville 2025.* Program organizer and mentor

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS) – *Portland 2023.* Panelist in a session on Multi-messenger Astronomy.

Lamat Program – *Santa Cruz 2023.* Mentor for Undergraduate Summer Student.

EMIT Summer School in Multimessenger Astronomy – *Vanderbilt 2023.* Developed tutorial notebooks for parameter estimation of gravitational waves.

Conferences for Undergraduate Women in Physics (CUWiP) – *Santa Cruz 2023.* Volunteer

Gravitational Wave Astronomy Northwest meeting (GWANW) – *Ligo Hanford 2022.* Lisa tutorial

Schrodinger's Pint – *Austin 2022.* Public lecture "Black holes and where to find them"

Astronomy on Tap talk – *Austin 2020.* News section

Nerd Nite – *Austin 2020.* Public lecture about gravitational waves.

Black Hole Night – *Austin 2019.* Hosted event to promote collaboration between local artists and scientists on topics relating to black holes.

CASA BAGRE - *Lima 2019.* Talk on Colliding Black Holes ("Agujeros Negros Collisionando") in collaboration with audiovisual artists.

Astronomy on Tap talk – *Austin 2017.* Public lecture about cosmic inflation.

Girl Day – *2019.* Volunteer for a UT event focused on exposing girls to STEM.

Alice in Wonderland volunteer – program focused on introducing high school girls to physics research.